

Dayi Fan

PERSONAL INFO

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EDUCATION

- 2022.01-CURRENT Ph.D. candidate in Computer Science and Engineering,
The Ohio State University, U.S.
Advised by **Prof. Xiaodong Zhang**
Topic: *Parallel Algorithm Design and its Hardware Acceleration
for Irregular Datacentric Applications*
- 2021.01-2021.05 Undergraduate Exchange in Computer Science,
National University of Singapore, Singapore
Advised by Prof. Trevor E. Carlson
Project: LSTM implementation with spiking neural networks
- 2017.09-2021.07 B.Eng. in Microelectronics Science and Engineering,
Southern University of Science and Technology, China
Advised by Prof. Quan Chen
Project: RRAM-based neural network optimization

PUBLICATIONS

- 2025 X-Blossom: Massive Parallelization of Graph Maximum Matching [[PDF](#)]
Dayi Fan, Rubao Lee, Xiaodong Zhang
*The 51st International Conference on Very Large Databases (VLDB 2025),
London, U.K.*
- 2024 X-TED: Massive Parallelization of Tree Edit Distance [[PDF](#)]
Dayi Fan, Rubao Lee, Xiaodong Zhang
*The 50th International Conference on Very Large Databases (VLDB 2024),
Guangzhou, China*

- 2024 RR-Compound: RDMA-Fused gRPC for Low Latency, High Throughput, and Easy Interface [\[PDF\]](#)
Liang Geng, Hao Wang, Jingsong Meng, **Dayi Fan**, Sami Ben-Romdhane, Hari Kadayam Pichumani, Vinay Phegade, Xiaodong Zhang
IEEE Transactions on Parallel and Distributed Systems (TPDS 2024)

SUBMITTED WORK

- 2026 X-Wim: Massive Parallelization of Weighted Matching in Bipartite Graphs
Dayi Fan, Simon Zhang, Rubao Lee, Hanqi Guo, Xiaodong Zhang
- 2026 Performance Insights and Implications of Traversal-based Graph Analytics on both Multi-core CPUs and GPUs
Jiaxin Liu, **Dayi Fan**, Rubao Lee, Cathy Xia, Xiaodong Zhang

TEACHING EXPERIENCE

- 2022.01-2024.05 Graduate Teaching Assistant
CSE 3421 Computer Architecture
The Ohio State University

ACADEMIC PERFORMANCE

Postgraduate GPA: 3.951/4.000
Undergraduate GPA: 3.90/4.00

- 2021.06 Excellent Graduate Award
Southern University of Science and Technology
- 2017-2020 First Class Student Scholarship (Top 5%, three consecutive years)
Southern University of Science and Technology

ADDITIONALS

Languages: English, Chinese, Cantonese

Music: Cello and Piano

Enthusiast: Photography and Cooking



March 30, 2026

To Whom It May Concern,

Mr. Dayi Fan is a Ph.D. candidate who will graduate soon, and I am his dissertation advisor. I am pleased to offer my strong support for his selection as a Shadow PC member for VLDB 2027. Dayi is highly qualified to serve in this role for the following reasons:

1. Dayi has tackled several classical and highly challenging graph problems, including Tree Edit Distance (TED), maximum matching via the Blossom algorithm, and the Hungarian algorithm for the assignment problem. He has developed efficient parallel algorithms and implementations on multicore CPUs and GPUs, demonstrating strong intellectual capability and technical expertise.
2. His Ph.D. dissertation is based on five publications in top venues: PVLDB (2024, 2025, 2026), IEEE Transactions on Parallel and Distributed Systems, and SIGMETRICS. Through this work, he has gained a deep understanding of the standards and expectations of leading research venues.
3. Dayi has helped me in reviewing submissions for conferences such as ICDE (2024 and 2025), PPOPP, and SoCC. He consistently demonstrates high-quality judgment, thoroughness, and a strong sense of responsibility, making him a reliable and effective reviewer.

If you have any questions, please feel free to contact me (zhang@cse.ohio-state.edu).

Sincerely,

Xiaodong Zhang, Ph.D.

Distinguished University Scholar and Robert M. Critchfield Professor in Engineering
Department of Computer Science and Engineering